

From Toy to Tool: Examining Student AI Use & the Effects of AI Scaffolding on Critical Engagement

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Abstract—As large language models (LLMs) become increasingly prevalent in educational settings, examining their perils and promises for underrepresented minority (URM) students remains critical. Our prior work showed that unscaffolded AI use in a culturally responsive computing (CRC) classroom was largely unproductive and created distractions to course-related learning. This follow-up study examines how intentional AI scaffolding within the AVELA - A Vision for Engineering Literacy & Access (AVELA) CRC framework fosters more critical and meaningful AI engagement among URM learners.

In this study, ChatGPT-3.5 was intentionally scaffolded into instruction as a learning aid, in lesson plans designed to encourage critical discussions around AI bias and limitations. In this approach, we examined student use cases and perceptions of AI within a structured and reflexive learning environment.

ChatGPT prompt log analysis, surveys, interviews, and classroom observations were used to examine how students engaged with AI and how they perceived its role in their learning. With over 70% of prompts categorized as “On-Topic”, findings showed significantly more task-focused engagement with AI scaffolding compared to the previous study without scaffolding. Although there are dependency concerns with introducing AI in classrooms, all students during observed conversations in class and their interviews consistently expressed awareness of AI’s potential drawbacks and possible negative impacts on creativity. Students reported using ChatGPT to support project completion, troubleshoot technical issues, and clarify course concepts, while also critically reflecting on AI’s efficacy, biases, and limitations.

Together, these findings suggest that scaffolded AI integration in CRC can transform LLM use from a toy-like distraction into an effective learning support tool while creating space for students to critically engage with the technology. More broadly, this work advances equity-centered approaches to AI literacy in computing education by demonstrating how intentional scaffolding can support productive and reflexive AI use among URM learners.

Index Terms—Engineering Education, Large Language Models, Software tools, Culturally Relevant Pedagogy

I. INTRODUCTION

As AI tools are becoming more common and accessible in personal, educational, and technical settings, students need sufficient support to leverage these new tools effectively. For underrepresented minority (URM) learners, especially those who identify as Black and Latine, barriers include not only access to technology, but also lack of culturally supportive learning environments and mentors [1], [2].

Studies have shown that culturally-affirming computing environments that encourage students to reflect on their identities, communities, and experiences lead to more retention and success in the field [3]–[5]. Unfortunately, with only 14% of the U.S engineering workforce and 19% of STEM bachelor degree holders identifying as Black/Latine as of 2023, this creates a dearth of technically trained and culturally representative mentors to support the upcoming generation of STEM learners [3], [5]–[8]. This gap leaves many URMs unprepared or unable to engage in STEM [7], [9], [10], limiting these communities’ ability to advocate for change or access economic opportunities. This is why we are investigating how technically trained and culturally relevant URM mentors can leverage AI to assist with technical content during culturally responsive computing (CRC) courses to better stretch their teaching resources.

As AI tools show promise in helping novice programmers complete tasks [11], there lacks a streamlined solution for how to effectively integrate it in CRC to assist with the mentor gap. In our preliminary study, we engaged the AVELA framework and introduced AI as a technical aid, but did so without providing scaffolding. Students were allowed to use AI without formal guidelines on *how* to use it, similarly to how students are given other online tools like Google [12]. The goal was to investigate if students naturally adopt AI as a learning tool solely based on their out-of-class interactions with AI,

or if it became a distraction and hindrance to learning. It was seen that students using ChatGPT without scaffolding often produced off-topic queries and relied on human instructors for help, rendering it ineffective as a teaching aide. That study highlighted that simply making AI available does not guarantee productive usage or learning [13]. What remains unknown is how *scaffolded* AI use and AI literacy changes student engagement, learning behaviors, and perceptions in a CRC setting [13].

To advance this work, we scaffolded AI literacy into the same course framework to investigate the ways students engage with AI and to assess whether it continues to function as a distraction or becomes a meaningful tool that supports learning. AI scaffolding aims to mitigate unproductive use by equipping students with structured strategies for interacting with AI, shifting use from passive or exploratory behavior to intentional and goal-oriented engagement.

This study is part of a broader research initiative in multi-modal and multicultural learning contexts, where community assumptions or inaccurate guidance can disproportionately affect URM learners. The overarching project asks: *How do URM learners in a culturally responsive Arduino/embedded systems course differ in use cases and engagement when AI literacy is introduced to position AI as a tool for learning?*

Research Questions

Our central research questions are:

- 1) How does the introduction of AI scaffolding influence the ways URM students use AI tools in a culturally responsive computing (CRC) environment?
- 2) In what ways does the introduction of AI literacy impact student engagement and learning behaviors compared to unscaffolded AI access?

Our data contributes to the growing understanding of how AI can be meaningfully integrated into CRC education, particularly for URM learners to close the opportunity gap. Specifically, this work highlights how structured support shapes not only whether students use AI, but how they use it.

We find that access to AI alone is insufficient. Effective and task-focused use requires AI literacy and intentional scaffolding that introduces and supports students through productive, context-relevant use cases. Without this support, students may default to ineffective or misaligned interactions, reinforcing disengagement or dependency on instructors. Students must be guided to critically analyze outputs, refine prompts, and apply AI-generated outputs in ways that align with their learning goals and identities, which is particularly important in CRC, where students' identities and lived experiences are central to meaningful learning [14].

This work positions AI scaffolding as a key instructional strategy for bridging gaps in digital and informational literacy, ensuring that emerging technologies function as tools for empowerment rather than barriers to participation.

II. BACKGROUND AND RELATED WORKS

As AI usage in classrooms continues to grow, there remains a dearth of research on the use of AI tools in URM classrooms with respect to CRC and culturally responsive pedagogy (CRP). Previous work has delved into these topics more granularly. For example, much data is available on the effects of CRP in CS curriculum from kindergarten to post-secondary curricula [15], [16]; effects of equity and equality in access to AI tools in the classroom, or lack thereof [1], [17]–[19]; racial liberation and self-efficacy through AI and CS education [4], [5], [20]; and benefits of near-peer mentorship in core curricula [1], [3], [13], [21], [22]. However, research gaps persist on integration techniques and their effects on student exploration of AI tools, understanding of course material, and the effects of inquiry-based learning to increase student agency [23].

A. Culturally Responsive Pedagogy (CRP)

CRP seeks to ground research and subjects in their own lived experiences, culturally relevant histories, and knowledge to create a more supportive framework for learning [13]. As an expanded idea, CRC applies CRP to engineering, programming, and related fields [13], [24]. CRC describes teaching techniques that are content-oriented and both relevant and responsive to students' cultural backgrounds and lived experiences. One such technique, called "Liberatory Computing," aligns activism skills, racial justice, and critical consciousness with computing education [5]. Under this teaching strategy, computing becomes more than just a technological skill, it expands into a tool for social transformation. Few studies examining the intersection of CRP in CS education have done so with a close examination of AI's role in the classroom [19], [22], despite the current data showing that upwards of 80% of schools use AI in their classrooms [25], [26].

B. Digital Neutrality and Bias Training

The digital divide is exacerbated in CRP and CRC by the presence of URM students who wish to learn and a systemic lack of URM educators in STEM [13], [27], who are vital for identifying and mitigating technological biases. While studies have shown that AI tools such as LLMs can assist novices in programming [11], work still remains to understand how technically inexperienced instructors can safely leverage these tools without inadvertently propagating embedded algorithmic biases. Despite the prevalence of AI in classrooms, students continue to gravitate towards human instructors, rendering AI unhelpful as a standalone instructor for technical content [13]. This gap between technically qualified, culturally aware instructors and biased AI tools has yet to be bridged [22], [26]. A clear need for multi-modal teaching assistance persists; however, the absence of quality training necessary to identify AI bias and bridge this divide remains one of the largest barriers to equitable STEM education [13], [17], [28].

Much has been studied on algorithmic bias and its effects on education [29], and other studies have looked at user biases and its influence on AI usage and perspectives [30]. It is

difficult to teach programming and critical interaction with AI software when the technology itself is viewed as a “neutral” tool [18] despite overwhelming evidence that it is not [29], [31]. In one review, the authors found at least four instances of AI-predictive behavior projecting less favorable outcomes for Black, Latine, or female students [29]. In addition, it is difficult to teach AI biases when the instructors themselves are novice users. Very few educators and school leaders report receiving adequate, regular, and comprehensive AI and AI bias training [25], [26]. In a national study published in 2025, 65% of teachers received AI literacy training, and of that only 35% reportedly found it helpful [25]. In the same study, an alarmingly low 14% of school leaders reported providing ethical and appropriate AI training to their students [25].

C. Digital Tools in the Classroom

As AI use increases in classrooms, more studies have tried to understand how it can be used to increase learning gains and how to best implement new digital tools. One review found that in over 75% of cases, digital tools had a positive effect on academic performance, student autonomy, and student engagement [32]. The important findings of this study are that storytelling using digital learning tools encouraged students to feel ownership, connection, and deep engagement with the material [32]. Other tools, called Search as Learning (SAL) tools, require students to engage purposefully, iteratively, and analytically in order to make use of them [12]. Examples of these types of SAL tools include web search engines and ChatGPT. With these technologies, it is useful to study what happens during the search process, and to determine how and to what extent the searching for information becomes the learning itself. One study shows that while scaffolded search functions may not increase learning gains, the use of scaffolded technology influenced the user’s search behavior [12]. Specifically, users became less focused on learning from the search results and more focused on filling in a gamified “progress bar” on the search engine itself [12]. We examine the same influence of scaffolded generative technology on user behavior and learning outcomes. Other research efforts have explored how interactive SAL tools, specifically ChatGPT, enhance the learning environment by tailoring responses to an individual student’s requests and needs [31].

D. Tool or Toy

Existing research on high school students and AI learning focuses their perceptions of AI, morality, equality, equity and accessibility [13]. However, less is known of the effects of the introduction of AI-literacy *on learning* in CRC settings [15], [19]. In our preliminary study, we examined how students adopted ChatGPT-3.5 in their learning without formal scaffolding. ChatGPT-3.5 was introduced in a culturally responsive embedded systems course over the summer within the AVELA framework, but without any explicit instruction on how to use it. We found that while initial interest in ChatGPT-3.5 was high (89% of students agreed/strongly agreed with being interested in AI), over the course of

the program that number decreased to 67%. Interestingly, 50% students reported an initial confidence in engaging with the software and 83% reported confidence in engaging with their near-peer instructors. By the end of the course, student confidence in using ChatGPT-3.5 had decreased to 67% while confidence in engagement with instructors rose to 92%. From this we learned that despite the initial excitement of using a new technology, students gravitated towards the human instructors, rendering AI not as helpful as a standalone instructor for technical content. When introduced as just another classroom tool, students used ChatGPT-3.5 in a variety of off-topic ways. Our study found that though 63% of student prompts pertained to coding of some sort, 50% of all prompts were off-topic. There was a stark disconnect between student engagement in the course subject and the topics most queried with ChatGPT-3.5. Students probed ChatGPT-3.5 to better understand ChatGPT-3.5 sensibility, with queries like, “what are you,” and “ask me a question” [13]. 34% of all prompts were categorized as Idea Expansion, meaning students used ChatGPT-3.5 for high-level exploration of class topics or personal interests. However, expansion on a class subject did not always translate to follow up, as some students did not integrate their findings into their final projects. Without intentional scaffolding or AI-literacy, students use ChatGPT-3.5 as a distraction or toy more than a tool to help their engagement with class content.

III. METHODS

A. Positionality

Our team is primarily composed of URM researchers. Five identify as Black and one identifies as Asian. Two of the researchers are women, and the remaining four are men. At the time of writing, all authors are active members, contributors, or partners of the AVELA nonprofit through which the class was taught. As people affiliated with predominantly White institutions, we are attuned to the systemic inequities facing URM students in computing. Many of us have been both experienced educators and technically inexperienced trainees under the AVELA framework and are active users of AI/LLMs, which informs our dual perspective as both practitioners and researchers. We note that while several authors have prior experience teaching similar student populations, only two authors were directly present in the classroom and observed the participants in this study. The remaining authors maintained complete distance from the classroom, supporting a more balanced interpretation of the data. All but two of the members of the research team hold university degrees in computing or engineering-adjacent fields. We acknowledge that our connection to AVELA and interactions with LLMs may shape how we interpret data and frame conclusions; however, we believe our positionalities offer culturally informed insights and strengthen the study’s relevance. By naming these positional dynamics, we aim to promote transparency and help readers better understand our interpretive lens.

B. Study Purpose

This study examines the learning behaviors and AI use cases of URM students in a culturally responsive embedded systems course with ChatGPT-3.5 introduced as a semi-structured, openly accessible classroom tool. The 40-hour program was conducted over 10 summer sessions at a career readiness program hosted by a local partnering non-profit community organization. Using 40 hours of embedded systems content from the culturally responsive AVELA framework [1], [3], the course objective was for students to design and build a prototype of a solution to fix a problem they have witnessed in their local community. The classroom structure used short instructional lessons on hardware concepts like sensors, user interface devices, digital and analog inputs, and basic circuits involving power, resistors, and LEDs. These lessons also covered topics about algorithmic bias for critical engagement with the AI tools at their disposal. Combined with group work, students iteratively brainstormed and developed their projects before presenting their final prototypes, including their hardware and code, in a formal presentation format. The students were provided with all necessary components and tools to complete and present their final projects, including laptops with new ChatGPT-3.5 accounts, Arduino tool kits, sensors, breadboards, display screens, etc.

Additionally, instructors provided scaffolding for the use of ChatGPT-3.5 in the course, including lessons that introduced students to the basics of LLMs and how ChatGPT-3.5 generates responses. They also guided students on how to effectively prompt ChatGPT-3.5 and interact with it to receive an expected output. Its use was non-restricted and available for students as a resource in addition to their peers, course instructors, and other non-AI online tools. Students were not explicitly encouraged to use any one tool over the other to allow us to explore how and when they naturally engage with ChatGPT-3.5. Students were only explicitly told to use ChatGPT-3.5 during AI literacy lessons.

The course engaged 24 URM secondary students and 4 near-peer instructors, 2 with technical experience with Arduino/embedded systems courses (experienced instructors) and 2 with minimal technical experience (inexperienced instructors). Participants were informed that a non-facilitating researcher would be observing the class to take notes on their progress and activities. They were instructed to ignore them and treat the study as a basic classroom [33]. Students were recruited locally through neighboring nonprofits and afterschool organizations via email blasts or word-of-mouth. This study was approved by the University of Washington's IRB. Written consent and assent were obtained from all participating students and parents prior to data collection.

C. AVELA Framework and Mentorship Model

The AVELA framework provided culturally relevant instruction, primarily by framing the course content to allow students to explore how their own identities shape their approach to their final projects. For example, students in the class were asked to use ChatGPT-3.5 to create a

fictional story about a character in STEM. This shared, affinity-based community space fosters a culture of reflexivity [4], empowering participants to critically engage with AI tools and course content in a way that directly reflects their own backgrounds and identities. We also note that while the AVELA model focuses on engaging URM students in STEM, this course welcomed students of all backgrounds.

Peer-to-peer and near-peer mentor-to-peer conversations were intentionally integrated throughout the course to enhance student engagement with the material [21], [34], [35]. This tiered mentorship approach has been shown to increase students' sense of comfort and belonging in STEM environments, particularly for those from URM backgrounds [36]. Because it is beneficial for both mentors and mentees, this structure played a critical role in shaping an inclusive and supportive classroom environment.

D. Surveys and Interviews

Surveys were administered before the course, weekly after each session, and after the culmination of the class to assess how AI-scaffolding influenced students' engagement and learning behaviors in the classroom. 27 students responded to the pre-course survey, 11 students on average responded to the weekly surveys, and 9 students responded to the post-course survey. The surveys included both open-ended and Likert-scale questions (1 = Strongly Disagree to 5 = Strongly Agree) and were structured around the same key themes as the interviews. Students were also asked to reflect on the role of AI in relation to fairness in society, with many responses highlighting themes aligned with liberatory computing [5]. This data was used to evaluate how students' experiences and perceptions of AI in the classroom evolved over the course.

After the course ended, researchers conducted interviews with all instructors and consenting students to gain insight into their experiences and perspectives on students' use of AI in the classroom. The interviews were organized around five core themes: demographics and background, perceptions of AI and ChatGPT-3.5, engagement with ChatGPT-3.5 in the course, self-efficacy and aspirations, and barriers and inclusivity. Each interviewee was asked a consistent set of questions across these categories, with interviews lasting approximately one hour. All interviews were recorded for research purposes with written and informed consent, and participants agreed to have their responses recorded, analyzed, and used in this study.

E. ChatGPT-3.5 Log Analysis

To better understand how students engaged with ChatGPT-3.5 during the course, we analyzed the prompts from the ChatGPT-3.5 logs of 24 students. The analyzed prompts were written independently by the students without instructor assistance, and instructor-assisted prompts were omitted from this study to focus specifically on student use cases. We analyzed the ChatGPT-3.5 logs through both consensus-based inductive thematic analysis and deductive analysis using codes from our previous study [13], [37]–[39]. The first researcher reviewed all student prompts and generated initial codes based

on patterns that emerged from the data. A second researcher, blind to the first coder's assignments, independently applied the same coding framework to the dataset.

Following this initial round, both researchers met to review and refine the codes and definitions, during which each researcher revisited their own coding while having visibility into the other researcher's assignments, making revisions as needed. After this independent refinement, the researchers met again to discuss discrepancies and reach consensus on the final codes. Inter-rater reliability was calculated using Cohen's Kappa [40] to assess agreement across themes and come up with percentage rates as shown in Fig. 1. This process resulted in 13 unique themes. Each prompt was assigned a minimum of two themes: On Topic/Off Topic, and Displays Prompting Technique/Displays Lack of Prompting Technique. Additional themes were added as necessary, and a manual Google spreadsheet was used to log and organize the data.

IV. RESULTS

We present our findings by examining how AI scaffolding and literacy training influence student AI utilization, engagement, and learning behaviors within a CRC environment, particularly in comparison to unscaffolded AI access. To evaluate these dynamics, we analyze multiple streams of empirical evidence, including qualitative and descriptive comparisons of ChatGPT-3.5 interaction logs, student surveys, and classroom observational data across both cohorts.

A. Final themes and definitions:

The ChatGPT-3.5 logs were categorized into thirteen distinct themes for student AI usage. These themes are listed below, with Fig. 1 illustrating the Cohen's Kappa inter-rater reliability and prominence of each theme.

- 1) **On Topic:** Prompts relating to class content.
- 2) **Off Topic:** Prompts unrelated to the assigned task or class content.
- 3) **Displays Prompting Technique:** Prompts demonstrating effective prompting strategies to achieve desirable responses without the need for reframing.
- 4) **Displays Lack Of Prompt Technique:** Prompts that lacked depth or framing to effectively achieve a desired response.
- 5) **Image Generation:** Prompts used to generate, edit, or request visual content such as images or diagrams.
- 6) **GPT Sensibility:** Prompts used to understand ChatGPT-3.5, such as questions about its sensibility, morality, and general functionalities.
- 7) **Coding Prompt:** Prompts related to students' code, either through questioning or copying and pasting their code directly into ChatGPT-3.5.
- 8) **Software Prompt:** Prompts pertaining to Arduino software rather than the coding or hardware aspect of projects.
- 9) **Hardware Prompt:** Prompts revolving around hardware and circuitry, both for class-related and non-class-related topics.

- 10) **Follow Up:** Prompts where students used GPT to elaborate on or follow up on personal or class-related concepts.
- 11) **AI Work Substitution:** Prompts used to delete entire task execution to ChatGPT-3.5.
- 12) **Reliance on ChatGPT for Ideation:** Prompts using ChatGPT-3.5 as the primary source for generating initial ideas, brainstorming, or defining direction and scope rather than building from independent thoughts.
- 13) **Tool for Learning:** Prompts used to clarify concepts, interpret errors, define terms, or explain foundational information to support understanding.

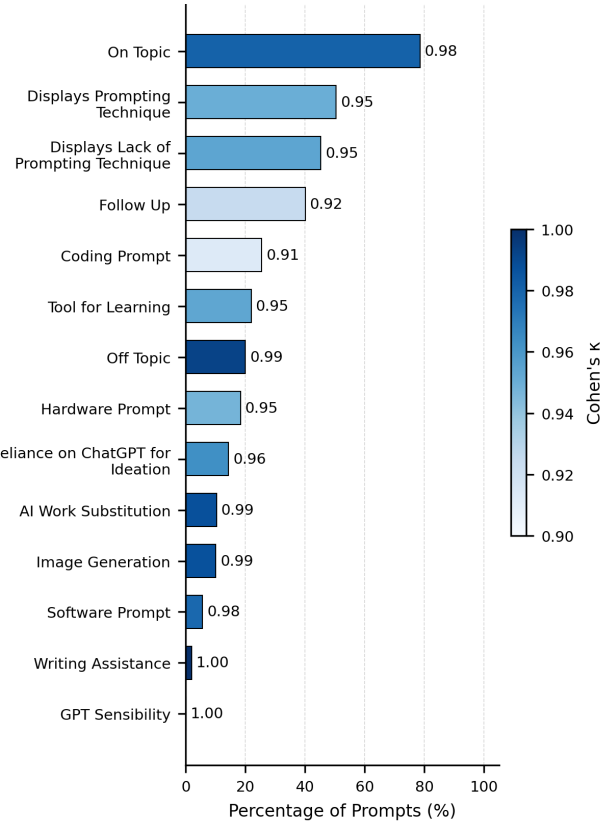


Fig. 1. Frequency of coded themes across student-generated ChatGPT-3.5 prompts in the AI scaffolding cohort, with color indicating the level of researcher agreement.

B. RQ1: AI Use Cases and Patterns

We were motivated to understand students' shift in AI use case patterns between the previous unscaffolded cohort who did not receive any AI literacy learning and the scaffolded cohort who did receive AI literacy learning. Students in the scaffolded condition demonstrated substantially higher rates of on-task AI usage compared to the unscaffolded cohort. Fig. 2 shows the difference between on-topic and off-topic queries between the two cohorts. In the scaffolded cohort, 79% of prompts were classified as on topic, compared to approximately 50% in the unscaffolded cohort from the preliminary study [13].

While both groups showed similar rates of AI use as a direct learning tool for course-related queries (20% for scaffolded cohort vs. 19% in the unscaffolded cohort), the primary

difference emerged in overall task focus. This suggests that scaffolding may influence how consistently students remain aligned with learning goals rather than whether they use AI at all. The cohort with scaffolding exhibited fewer off topic and low-value interactions, including personal or gaming questions or responses to test ChatGPT-3.5's sensibility, which decreased from 22% with the group without scaffolding to less than 1% in the group with scaffolding.

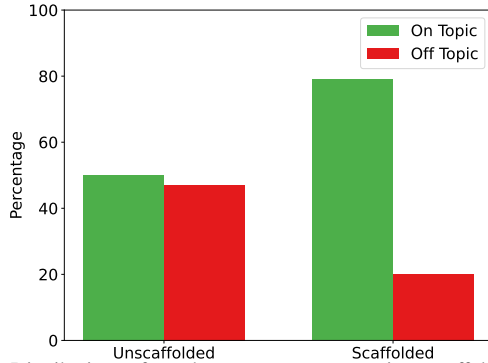


Fig. 2. Distribution of student prompts comparing scaffolded versus unscaffolded conditions.

1) *Depth Engagement with Course Material:* Another key difference between groups was the depth of engagement. Approximately 30% of student interactions in the cohort with scaffolding involved iterative follow up prompting for on topic queries, compared to 0% in the unscaffolded group for being a follow up *and* on topic. This suggests that while students without formal scaffolding accessed AI, they did not naturally use it to deepen understanding or extend their learning of course content through multi-step inquiry.

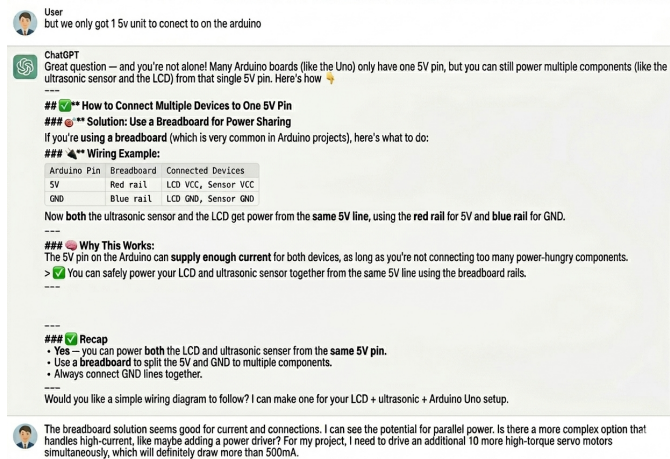


Fig. 3. An example of iterative follow-up prompting where a student engages in multi-step inquiry to troubleshoot a hardware constraint, where the first white section represents the student's initial message, the gray section represents the LLM's response, and the second white section represents the student's follow-up.

2) *Breadth of AI Use Cases:* Students in the scaffolded cohort used AI across a wider range of tasks when used for course related projects, including hardware (19%), coding (25%), software (6%), and final project assistance, including ideation and presentation script writing (15% combined).

Students in the cohort without scaffolding used the AI for 16% hardware, 28% coding, 0% software, and 0% for final project assistance.

Notably, only approximately 25% of prompts were categorized as work AI-substitution and Reliance on AI for Ideation combined, indicating that most interactions were grounded in active problem-solving rather than outsourcing thinking. This is consistent with survey data where pre and post-class surveys indicated that the majority of students do not strongly think of AI as increasing creativity (37% and 33% respectively) and students recognizing that AI is prone to mistakes, as indicated by 74% of responses pre-survey and 78% in the post survey.

C. RQ2: Impact on Engagement and Learning Behaviors

We examine how AI scaffolding influences student engagement and learning behaviors by comparing scaffolded and unscaffolded cohorts. Results indicate that students in the scaffolded cohort demonstrated higher levels of engagement, more effective prompting techniques, and increased use of AI for iterative learning and problem-solving. However, these benefits were accompanied by emerging challenges in classroom dynamics, particularly in student expectations of instructors and reliance on AI tools.

1) *Impact on Learning Behaviors:* The introduction of scaffolded AI in the classroom increased student engagement with classroom content and improved their abilities to interact with the LLM meaningfully. With scaffolding, students displayed proper prompting technique in 50% of prompts, which is significantly higher than the 13% demonstrated in the cohort without scaffolding. This means that the student used more intentional wording and context to provide clear, specific, and well-structured prompts that guided ChatGPT-3.5 toward generating relevant outputs when they had AI scaffolding.

Additionally, there was a notable descriptive difference in LLM engagement between the two classrooms. While students in the unscaffolded cohort averaged 3 prompts per student, those in the scaffolded cohort averaged 22 prompts per student. This higher volume of prompting suggests greater interaction with ChatGPT-3.5 within the scaffolded environment, though the lack of experimental control prevents drawing a direct causal link. This trend toward increased usage coincides with a positive shift in student perception regarding the tool's utility. In the pre-course survey, 85% of respondents indicated that ChatGPT-3.5 could help them understand computing concepts better, while 7% indicated that it could not. By the end of the course, alongside the observed increase in prompting, students' perceptions shifted slightly: the proportion of "Yes" responses rose to approximately 89%, and notably, no students selected "No" when asked "Do you think AI/ChatGPT can help you understand computing concepts better?"

2) *Instructor Observations:* In the scaffolded cohort, instructors often observed that students increasingly favored AI when seeking help quickly. This led students to expect similarly rapid responses from human instructors. During a post-class interview, one instructor noted,

Because they got used to asking ChatGPT-3.5, they think things should happen that fast.

– **Inexperienced Instructor 1**

The same instructor also described a pattern in the classroom where students were disengaged during instruction, anticipating that they could rely on AI tools during the activity. When they were unsuccessful in completing assigned tasks because of their lack of focus during the instructional period, the students grew frustrated with both the technology for not responding with the expected outputs, and the instructors for not being able to catch them up to speed as quickly.

So as an instructor, I feel like [ChatGPT-3.5] just added more stress to the environment. Whereas if students were learning authentically from the start, they would have then gained the knowledge needed to proceed through the finale.

– **Inexperienced Instructor 1**

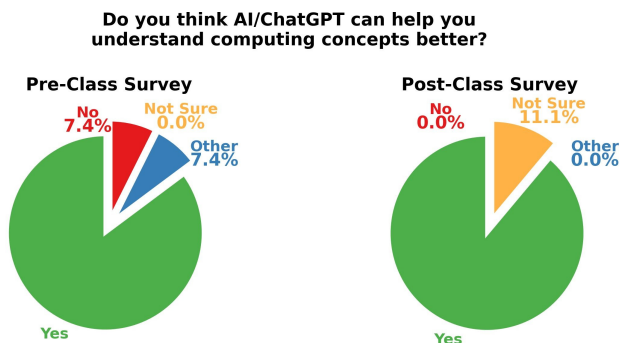


Fig. 4. Student responses regarding whether they believe ChatGPT-0.2 helps them understand complex concepts in the scaffolded condition.

3) *Independent Problem Solving and On-Demand Support:* Observational data indicated that students frequently used AI for debugging and iterative problem solving prior to seeking instructor assistance. This suggests a shift toward increased independence, where AI acts as an intermediary support before human intervention. 40.09% of all student entries were tagged as a 'Follow Up' to a previous inquiry. 32.35% of entries were follow ups that pertained to class content/were on topic. This highlights the behavior of students' independent iterative problem solving using AI.

Students were able to use ChatGPT-3.5 for 'Just-in-Time Debugging' to work through solutions independently before reaching out to instructors for final validation, citing that it's natural language abilities made them more comfortable to have dialogue about bugs. One student described using ChatGPT-3.5 to clarify course content and work through their confusion points so that they felt more prepared to ask detailed questions to the instructors.

"I learned how to word my questions better, like the prompt to the AI ... Like I knew how to word it better so that the instructor could answer".

– **Student 8**

There was also a difference in the types of questions students asked ChatGPT-3.5 vs the types of questions they asked their instructors. In student interviews, students highlighted that

they tended to ask the AI questions regarding coding and software, and reserved questions about the physical hardware for the instructors. Student ChatGPT-3.5 logs confirmed this, with approximately 25.5% of entries relating to coding and 18.4% relating to hardware.

Students' use of AI for technical content also allowed the culturally relevant yet technically inexperienced instructors the agency to be culturally relevant near-peer mentors who leveraged AI to assist with technically complex concepts, thus stretching their ability to be effective instructors during the course.

So if I didn't know the answer and all the other instructors were busy helping someone else, then me and that student would go to chat and try to troubleshoot together.

– **Inexperienced Instructor 2**

V. DISCUSSION

Our findings highlight the complex role that AI scaffolding plays in CRC environments. While prior work has often framed AI as either a disruptive force or a productivity aid, our results suggest a more nuanced reality: AI can function as a tool for transformative learning when its use is *intentionally scaffolded*, but may otherwise contribute to distraction, shallow engagement, or misaligned learning behaviors.

A. AI as a Tool for Transformative Learning

When supported through structured scaffolding, AI tools can extend beyond convenience-based use and instead support deeper engagement, iterative thinking, and both student and novice instructor agency. In our study, scaffolded cohort students demonstrated higher rates of follow-up prompting, more intentional prompt construction, and increased use of AI for debugging and concept clarification. These behaviors suggest that students were not only using AI to obtain answers, but to actively participate in their own learning processes [41].

This aligns with frameworks such as "I Am a Technology Creator: Black Girls as Technosocial Change Agents in a Culturally-Responsive Robotics Camp" [2], [42], [43], which emphasize shifting students from passive consumers of technology to active creators and critical users. Within this context, AI becomes not just a tool for efficiency, but a medium through which students can explore, refine, and construct knowledge in ways that are responsive to their needs and identities.

Importantly, students reported that AI made information more accessible, particularly through natural language interaction. The ability to ask repeated questions, request rewording, and engage without fear of judgment created a learning environment that some students found more approachable than traditional instructional settings [24]. This was particularly observed in cases where students' first language differed from the language of instruction, so the students turned to ChatGPT-3.5 to simplify concepts and make them digestible and easier to understand. In CRC

environments, where student identity, voice, and comfort are central, this form of interaction may be especially impactful.

B. Scaffolding Shapes Use, Not Just Access

A key finding is that AI scaffolding influences how students use AI, rather than whether they use it at all. While both scaffolded and unscaffolded groups demonstrated similar rates of AI use as a direct learning tool (20% vs. 19% respectively), the scaffolded group showed substantially higher levels of overall task focus and significantly fewer off-topic interactions.

This distinction is critical. Our results suggest that access to AI alone does not ensure meaningful engagement. Instead, without guidance, students may default to unfocused, exploratory, or low-value interactions. With scaffolding, however, AI use becomes more consistently aligned with learning objectives.

Given that students are likely to use AI tools regardless of instructional design, these findings position scaffolding as an essential pedagogical strategy. Rather than attempting to limit or control AI use, educators may be better served by explicitly teaching students how to engage with these tools productively, critically, and in ways that support their learning goals.

C. Causation vs. Correlation

While scaffolded cohort students demonstrated higher engagement, more on-task behavior, and increased use of effective prompting strategies, these findings should be interpreted with caution. The similarity in learning-tool usage rates suggests that scaffolding alone may not directly increase whether students use AI for learning purposes. It is also important to note demographic differences between cohorts such as prior experience with AI, or other contextual factors not controlled for in this study, may have influenced results. As such, we cannot definitively attribute observed differences solely to the introduction of scaffolding. Instead, our findings are best understood as correlational, indicating a strong relationship between scaffolding and more productive patterns of AI use. Future work is needed to isolate causal mechanisms and better understand how specific scaffolding strategies influence student learning behaviors.

D. Emerging Tensions: Efficiency vs. Learning

AI scaffolding supported engagement and independence, but also introduced new tensions. Instructors reported that students expected AI-like response times from human instructors, creating frustration when those expectations were unmet.

Additionally, some students disengaged during instruction, expecting to rely on AI tools later to compensate. This behavior contributed to a negative feedback loop, where limited initial understanding led to ineffective AI use, which increased students' reliance on instructors for clarification.

These findings highlight an important tradeoff: while novel educational tools like AI can accelerate access to information, it may also shift student expectations in ways that challenge traditional instructional pacing and interaction [12], [23]. Designing learning environments that balance immediacy with depth remains an open challenge.

E. Implications for Culturally Responsive Computing

For URM learners in CRC environments, the implications of this work are particularly significant. Prior research has shown that access to technology alone is insufficient to support meaningful participation in STEM. Our findings extend this understanding to AI, demonstrating that without intentional scaffolding, these tools may reproduce existing gaps in digital and informational literacy.

When thoughtfully integrated, AI can support more inclusive learning by enabling natural language interaction, iterative questioning, and on-demand assistance that may reduce barriers related to confidence, communication, and access to immediate assistance.

In this sense, AI scaffolding can be understood as a form of support for culturally environments: one that meets students where they are and provides flexible pathways for engagement. Ensuring that these tools are introduced with both technical and cultural intentionality will be critical to their success in broadening participation in computing.

VI. CONCLUSION

This follow-up work contributes to the growing body of knowledge on AI-assisted learning and CRC frameworks, underscoring the necessity of AI literacy and scaffolding prior to classroom tool integration. While our uncontrolled data groups limit our ability to isolate scaffolding as the sole driver of change, the observed descriptive trends suggest a positive association between intentional guidance and a student's ability to prompt effectively, remain on task, and exercise greater learning agency. Furthermore, AI literacy training appears to better prepare students to leverage these tools for personalized benefits, including English language support, on-demand assistance, and supplementing technical problem-solving with novice instructors.

Together, these findings suggest that scaffolded AI integration in CRC environments can help transition AI use from a toy-like distraction into an effective tool for task support, while providing students the structural space to critically engage with the technology. Ultimately, this work advances equity-centered approaches to computing education by highlighting that access to AI alone is insufficient without intentional guidance in its application.

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